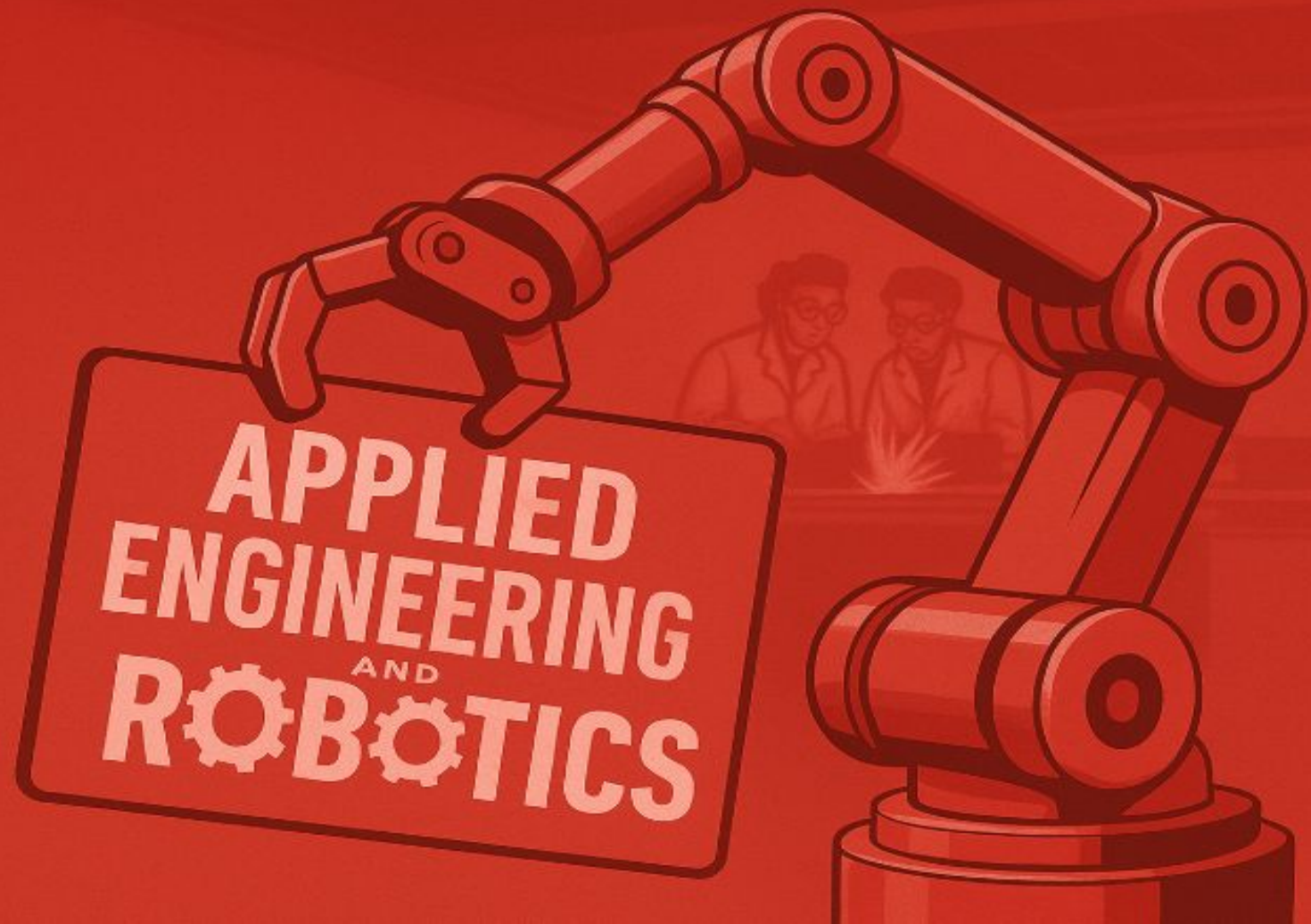




Available CAD Solutions

Desktop

- Autodesk Inventor
- SolidWorks
- CATIA
- PTC Creo
- Siemens NX
- Rhinoceros (Rhino)

Cloud

- Autodesk Fusion 360
- Onshape
- TinkerCAD

Open-Source

- FreeCAD
- Blender
- SketchUp

Learning Fusion 360 From Scratch: A Beginner's Guide



AUTODESK
Fusion 360

Getting Set Up - Fusion 360

Fusion 360 Accounts

- Open Fusion 360 desktop version and login with your school email

Safety Brief

No shop tools today—avoid sharp pencils

Keep workspace tidy (5S: Sort, Set in order)

Avoid kicking wires under the desk

Report any technical issues to instructor

Fusion 360 Interface Tour

Launch Screen:

- Overview: Welcome screen with recent files and new project options
- Action: Click "New Design from Scratch" to start

Main Workspace:

- System Tools (File menu, save, undo/redo)
- Canvas: 3D modeling area (center) where sketches and models appear
- Toolbar:
 - a. Top level tabs (e.g., Solid, Surface, Mesh)
 - b. Secondary level groups (e.g., Create, Modify, Constraints)
 - c. Tools (e.g., Line, Rectangle, Circle, and contents of dropdown)
- Browser: Left panel showing design history, bodies, and components

Fusion 360 Interface Tour

Navigation Tools:

- ViewCube: Top-right corner to rotate view (click faces or edges)
- Orbit/Pan/Zoom: Middle mouse button or toolbar icons for navigation
- Tip: Hold Shift + middle mouse for orbit

Timeline:

- Bottom area tracking design steps (editable history)
- Use: Click steps to revisit or edit

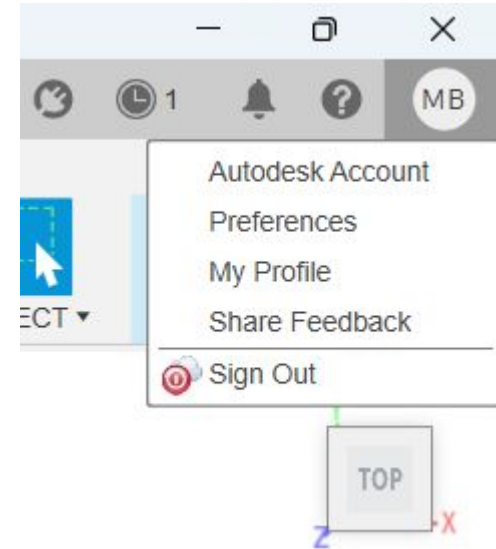
Quick Tips:

- Hover over icons for tooltips
- Use F5–F8 for view presets (e.g., F5 for top view)
- Ask for help if lost!

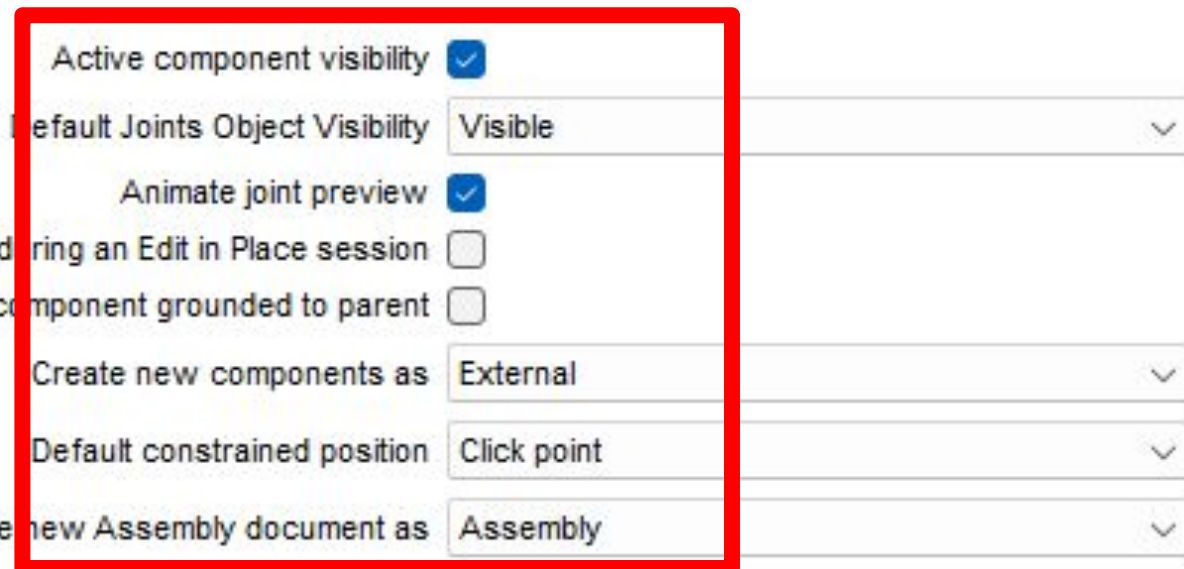
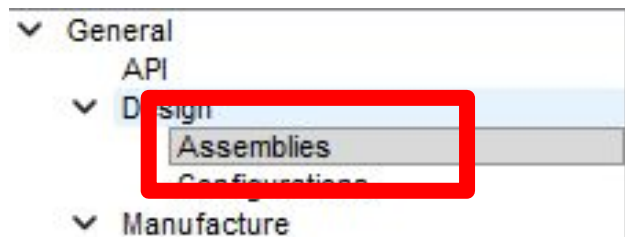
Fusion 360 Preferences

Click on your account settings icon, and select preferences

- Reverse Zoom Direction if desired
- Change Pan, Zoom, Orbit shortcuts if desired



Fusion 360 Preferences



Group 1	
Brady	Gibson
Joshua	Peet
Louis	DeVito
Group 2	
Lyla	Veldt
Zachary	Goss
Lukas	Staples
Group 3	
Domanik	Emmons
Milo	Hildebrant
Aiden	Deal
Group 4	
Ava	Swaney
Connor	Montgomery
Victoria	Hillard
Group 5	
Graem	Rey
Gavin	Phillips
Peyton	Slone
Group 6	
Avery	Moyer
Gage	VanCleave
Madelyn	Steigenberg
Keenan	Wilson

Group 1		
Jackson	Weiss	
Savannah	Ellis	
Timothy	Gustafson	

Group 2		
Zane	Davis	
Zoe	Roodbergen	
Landon	Krueger	

Group 3		
Graeson	Heggie	
Ashton	Laham	
Chris	Simon	

Group 4		
Quincy	Schwartz	
Gregory	Berndt	
Makenzie	Kleckner	

Group 6		
Nicholas	Simon	
Jorden	Stratton	
Mekhi	Stephens	
Kellan	Ferris	

Group 7		
Mason	Kreitner	
Logan	Frazier	
zachary	shannon	
Karsyn	Speelman	

Guide to Making Constrained Sketches

What Are Constrained Sketches?

- Sketches where geometry is fully defined (black lines in Fusion 360)
- Prevents unwanted changes with parametric editing

Why It Matters:

- Ensures accuracy in engineering designs
- Ties to EDP "Plan" step (precise planning)

Steps to Create Constrained Sketches

Step 1: Start a New Sketch

- Open Fusion 360 > Create > Sketch
- Select a plane (e.g., XY plane)

Step 2: Draw Basic Shapes

- Use tools: Line, Rectangle, Circle

Step 3: Apply Geometric Constraints

- Horizontal/Vertical: Align lines
- Parallel/Perpendicular: Match angles
- Tangent: Connect curves to lines
- Equal: Match lengths or radii
- Tip: Right-click to add constraints

Steps to Create Constrained Sketches

Step 4: Add Dimensions

- Click Dimension tool
- Set lengths, angles, or radii (e.g., 50 mm)
- Edit to test parametric changes

Step 5: Verify Full Constraint

- Look for black lines (fully defined)
- Blue lines indicate under-constrained areas

Example: Draw a rectangle with mounting holes for a cover plate

Example: Draw a rectangle with mounting holes for a cover plate

- Open new sketch
- Draw a rectangle
 - a. Add dimensions during draw or add them after
 - b. Start rectangle at origin to constrain to origin
- Select rectangle tool again and change line type to construction
- Dimension each line from the outer rectangle
- Cntrl + Z until dimensions and inner rectangle are gone
- Select Offset Tool
- Create offset from first rectangle of -10mm

Example: Draw a rectangle with mounting holes for a cover plate

- Place circle on one corner of inner rectangle, add dimension and hit enter
- Place circle on second corner. Use equal constraint to constrain size to other circle
- Cntrl + Z until only first circle remains
- Create > Rectangular Pattern
- Select circle, select line to indicate direction
- Enter distance (extent) and quantity (total including original)
- Press OK
- Repeat to make two more circles
- Cntrl + Z until only two circles remain
- Select line tool, select construction type
- Hover over midpoint of inner rectangle and create a centerline
- Select mirror tool, select circles, define mirror line

Example: Draw a rectangle with mounting holes for a cover plate

- Create a circle inside each corner of main rectangle w/o touching lines
- Create tangent constraint for each circle to each line of the corners
- Dimension each circle to 2mm or use equal constraint
- Use trim tool to trim corner off and inside of circle
- Cntrl + Z until circles are gone. Count steps
- Open fillet tool, select first corner, enter dimension (2mm), add remaining corners, hit enter

Tutorial Videos

[Fusion 360 in 30 Days Video Tutorial Series](#)

[Day 1, Lego Brick](#)

[Day 2, Soda Bottle](#); (Image is in Google Classroom, Unit 2; or download it from the comments section of the video)

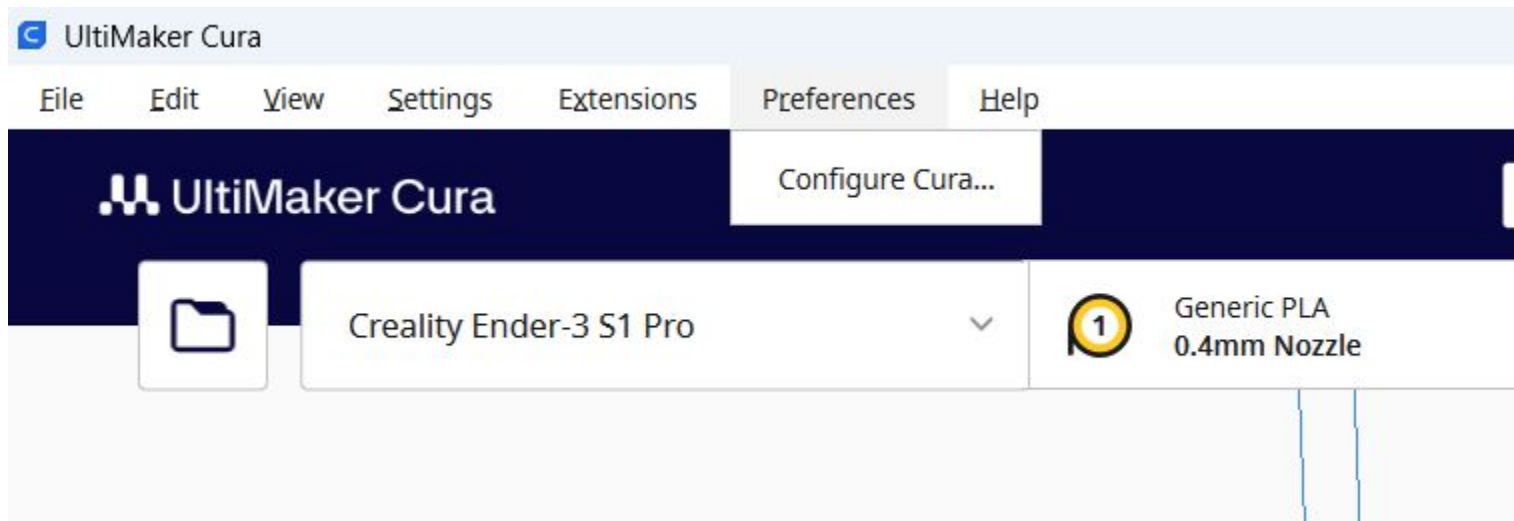
[Day 3, Paperclip](#)

[Day 4, Square Bottle](#)

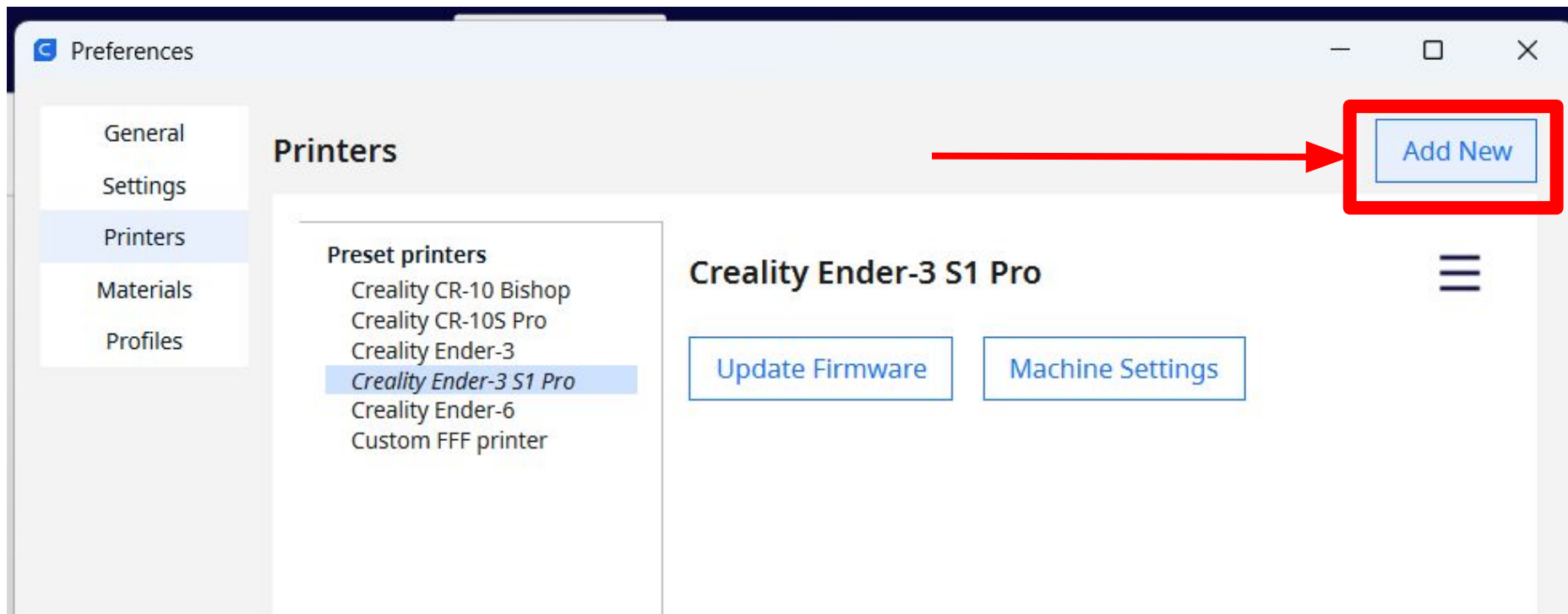
Ultimaker Cura

<https://www.youtube.com/watch?v=7lJmkTSoWLc>

Configuring Cura



Configuring Cura



Configuring Cura

✕ Add Printer

✕

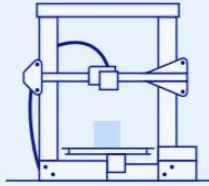
Add printer

In order to start using Cura you will need to configure a printer.

What printer would you like to setup?



UltiMaker printer



Non UltiMaker printer

1

2



Add a non-networked printer

Configuring Cura

Add a non-networked printer



ender-3 v3 KE

Add Printer



Add printer

Add a networked printer



Add a non-networked printer



ender-3 v3 KE



Creality3D

Creality Ender-3 V3 KE

Creality Ender-3 V3 KE

Manufacturer Creality3D
Profile author trouch.com
Printer name

Creality Ender-3 V3 KE

Add Ultimaker printer via Digital Factory

Add

Configuring Cura

Add Printer



Machine Settings

Creality Ender-3 V3 KE

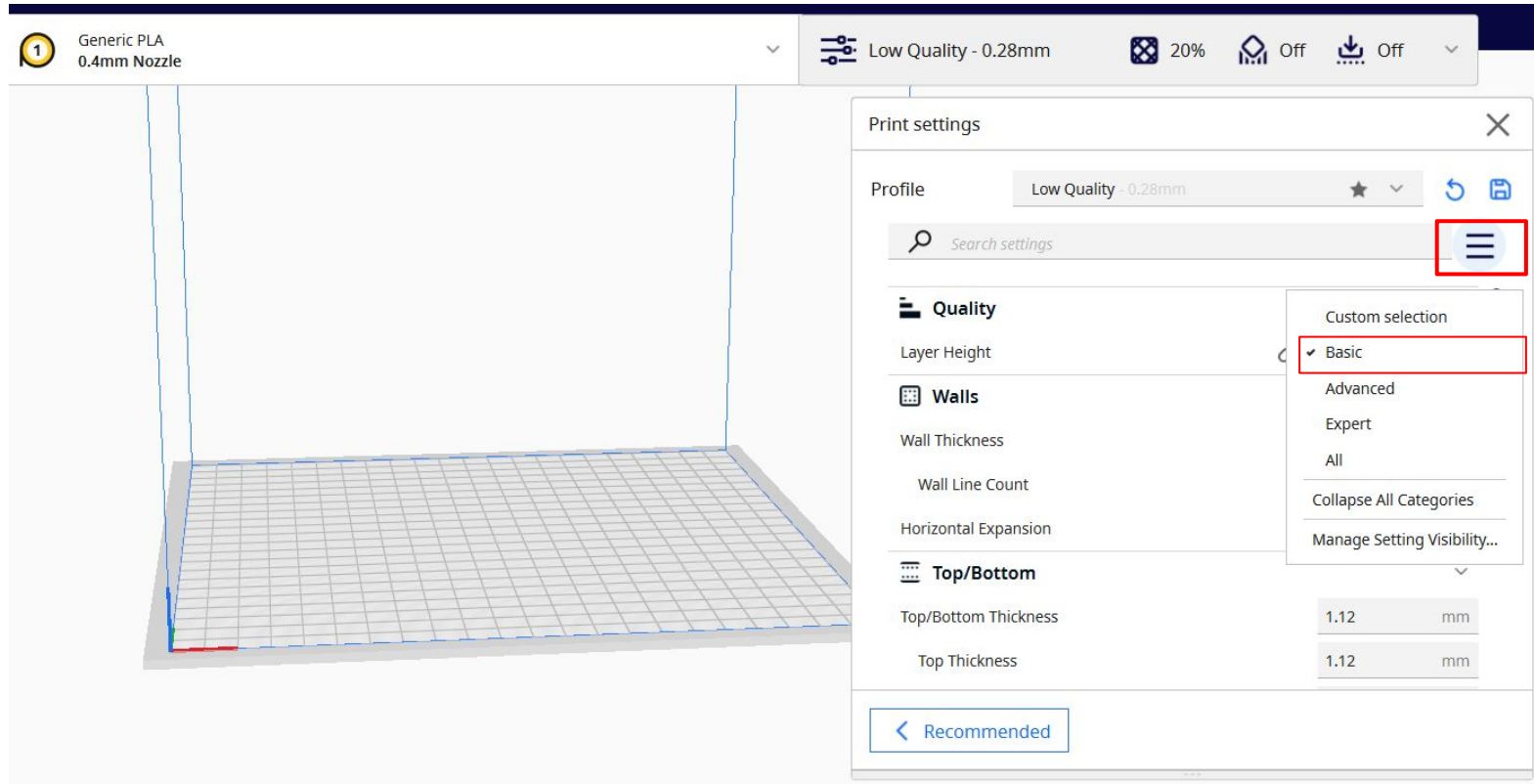
Printer		Extruder 1	
Printer Settings		Printhead Settings	
X (Width)	220.0 mm	X min	-20 mm
Y (Depth)	220.0 mm	Y min	-10 mm
Z (Height)	240.0 mm	X max	10 mm
Build plate shape	Rectangular	Y max	10 mm
Origin at center	☐	Gantry Height	38.0 mm
Heated bed	☒	Number of Extruders	1
Heated build volume	☐	Apply Extruder offsets to GCode	☒
G-code flavor	Marlin		
Start G-code		End G-code	
<pre>M220 S100 ;Reset Feedrate M221 S100 ;Reset Flowrate G28 ;Home</pre>		<pre>G91 ;Relative positionning G1 E-2 F2700 ;Retract a bit G1 E-2 Z0.2 F2400 ;Retract and raise Z G1 X5 Y5 F3000 ;Wipe out</pre>	

[Learn more](#)



Next

Configuring Cura - Select Basic Settings Visibility



Configuring Cura

 Quality 		
Layer Height		0.28 mm
 Walls 		
Wall Thickness		0.8 mm
Wall Line Count		2
Horizontal Expansion		0.0 mm
 Top/Bottom 		
Top/Bottom Thickness		1.12 mm
Top Thickness		1.12 mm
Top Layers		4
Bottom Thickness		1.12 mm
Bottom Layers		4

Configuring Cura

Infill

Infill Density

20.0

%

Infill Pattern

Cubic

Material

Printing Temperature

200.0

°C

Build Plate Temperature



60

°C

Speed

Print Speed

300.0

mm/s

Travel

Enable Retraction



Z Hop When Retracted



Configuring Cura

 **Cooling** 

Enable Print Cooling

☒

Fan Speed

f_x

100.0

%

 **Support** 

Generate Support

  ☒

Support Structure



Normal

▼

Support Placement



Everywhere

▼

Support Overhang Angle



35.0

°

Support Horizontal Expansion



0.8

mm

 **Build Plate Adhesion** 

Build Plate Adhesion Type



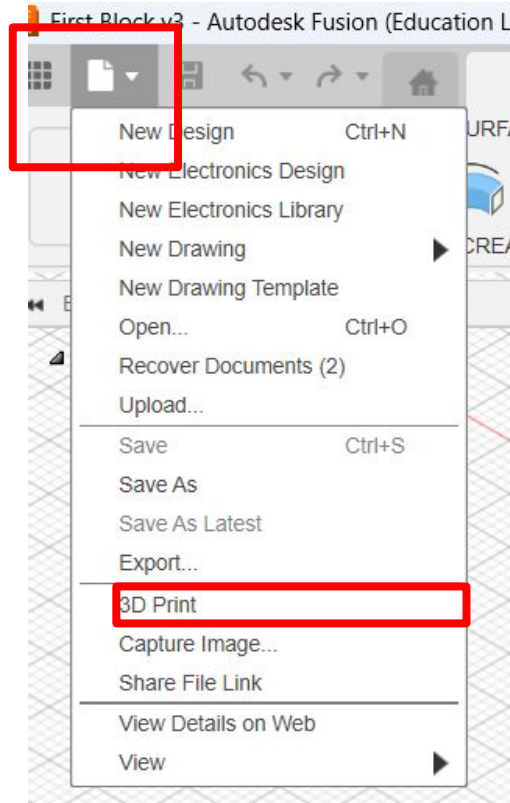
Skirt

▼

 **Dual Extrusion** 

 Recommended

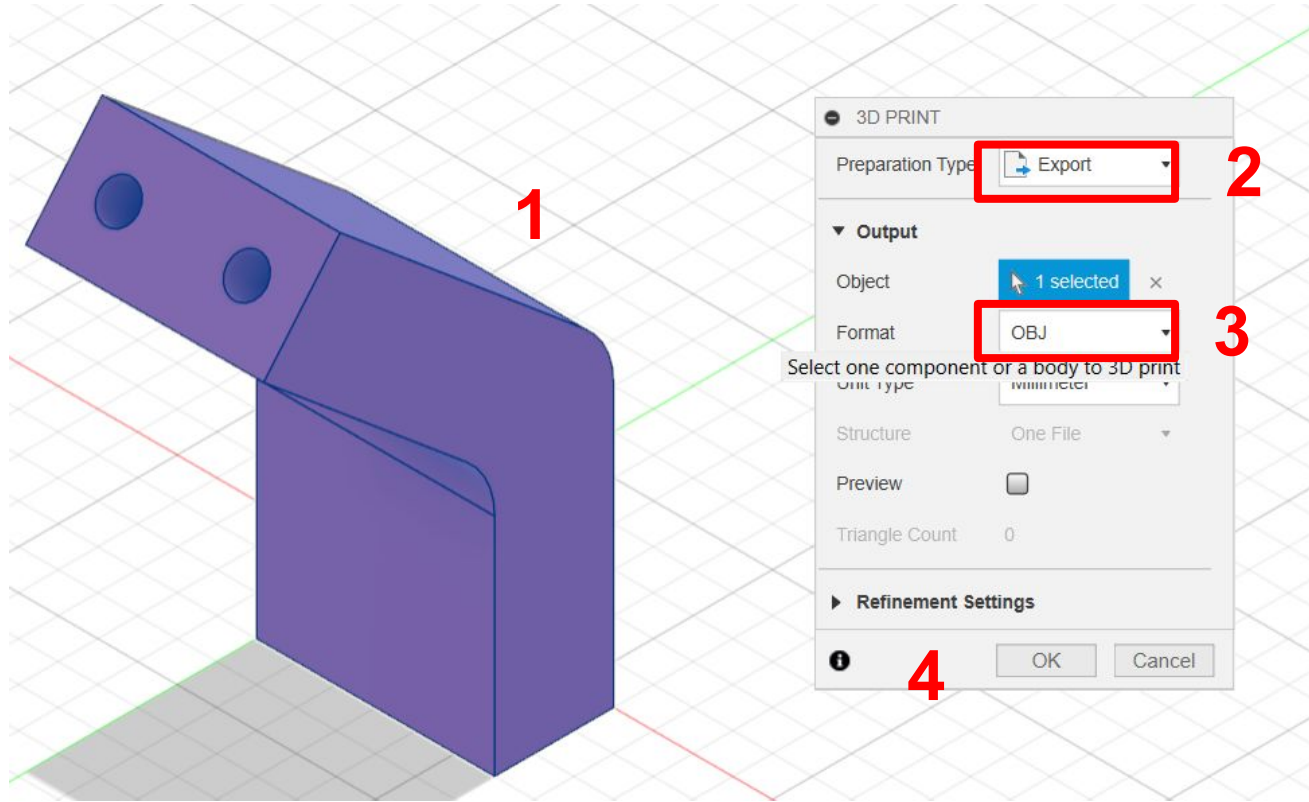
Export Model from Fusion 360 for 3D Printing



Select 3D Print from the File menu

Export Model from Fusion 360 for 3D Printing

1. Click on your model to select it.
2. Ensure preparation type is set to Export.
3. Select OBJ format.
4. Click OK. Set filename and path in next window.



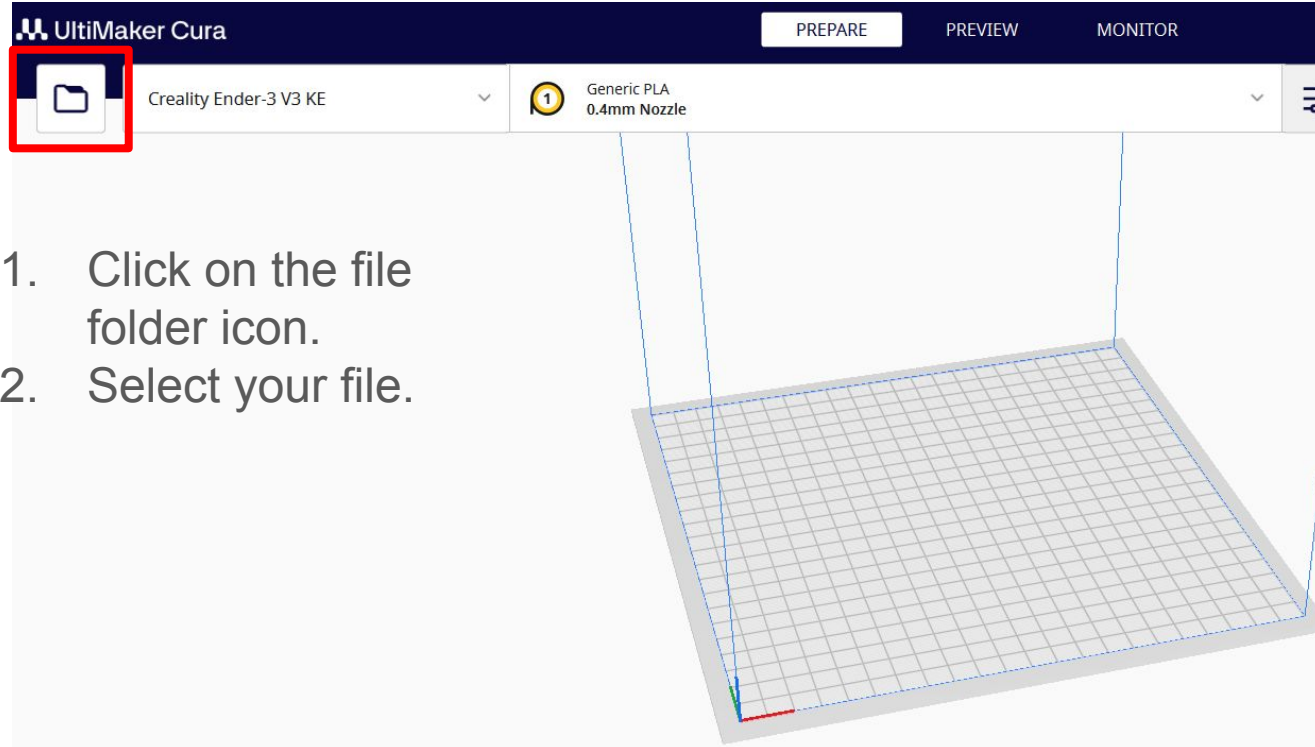
STL vs OBJ File Format

STL (Standard Tessellation Language) files store a 3D object's surface geometry using a mesh of triangles, making them ideal for simple 3D printing but lacking color or texture data.

OBJ (Wavefront Object) files are more versatile, supporting various polygon shapes, and can also include color, texture, and material information, though this requires a separate MTL file. The best format depends on the application: use STL for basic 3D printing, and OBJ for rendering, detailed geometry, or models with color and texture.

Importing File to Cura

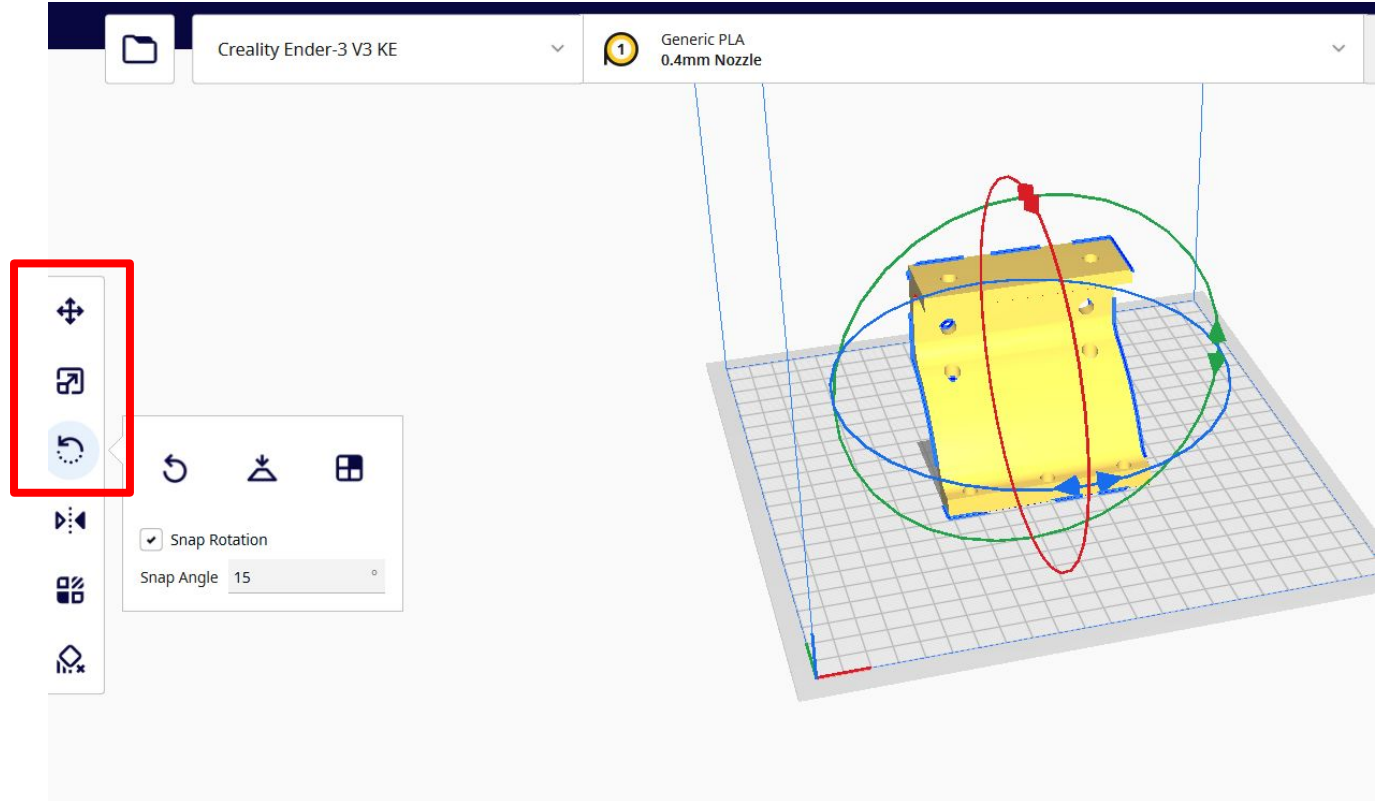
1



1. Click on the file folder icon.
2. Select your file.

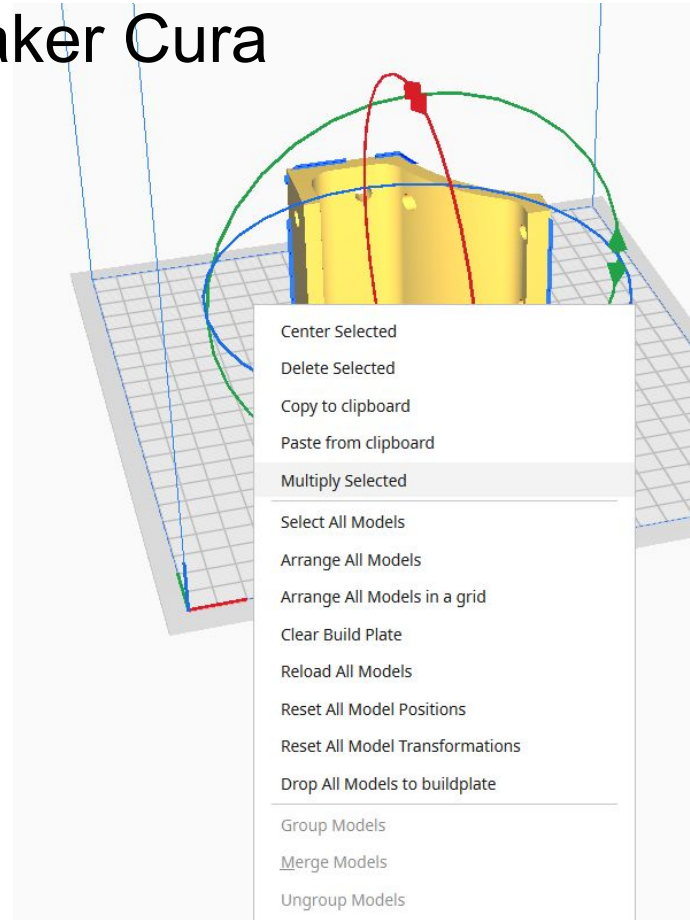
Slicing Parts for Printing in Ultimaker Cura

Use the toolbar on the left to activate tools to manipulate your model. Orient the model to optimize print strength, build plate adhesion, and to reduce the need for supports.



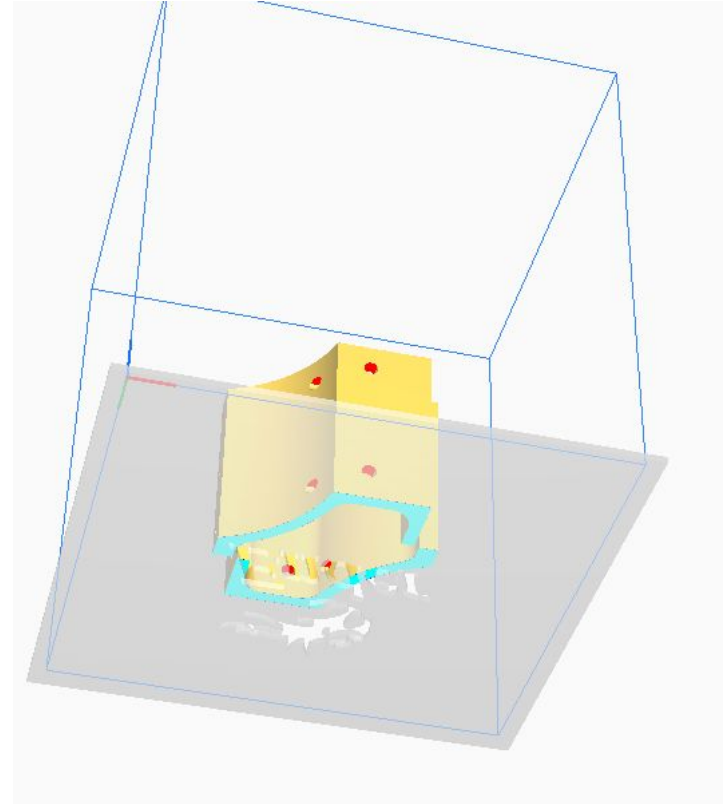
Slicing Parts for Printing in Ultimaker Cura

Right click on the build plate to access other options.



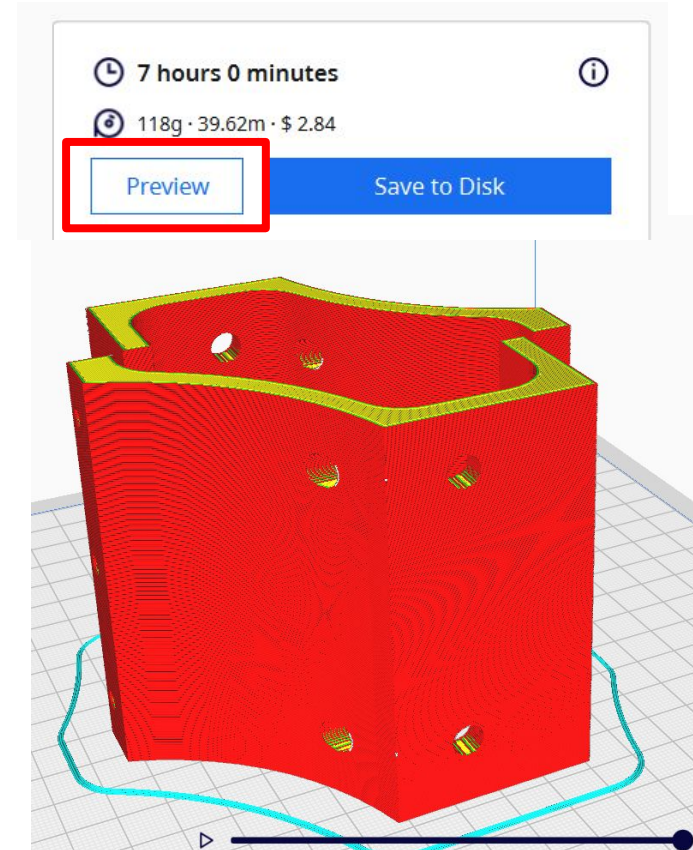
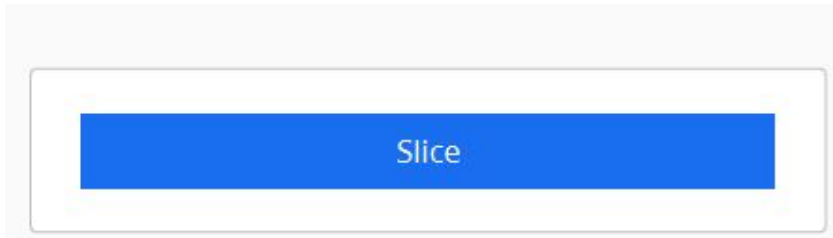
Slicing Parts for Printing in Ultimaker Cura

Right click and hold to rotate the model to examine all aspects. Ensure the surface touching the build plate is cyan. Red parts indicate areas that may require supports.

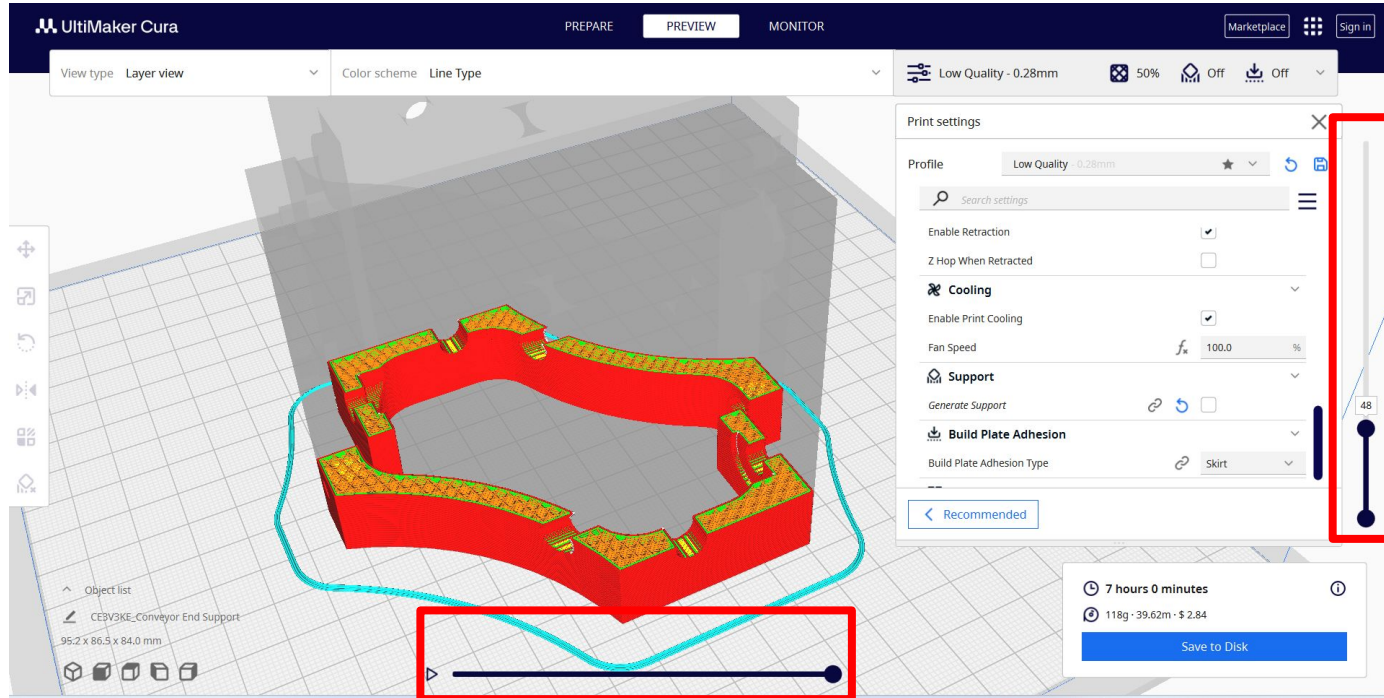


Slicing Parts for Printing in Ultimaker Cura

Slice the part and click on the preview button to examine the sliced part. Notice time and material estimates



Slicing Parts for Printing in Ultimaker Cura



Use sliders on bottom and right side to examine layers.

Slicing Parts for Printing in Ultimaker Cura

 **7 hours 0 minutes**



 **118g · 39.62m · \$ 2.84**

Save to Removable Drive



Insert thumb drive into your computer. “Save to Disk” changes to “Save to Removable Drive” after a moment. Click to save and remove your drive.

Tutorial Videos Week of 9-22-25

[Fusion 360 in 30 Days Video Tutorial Series](#)

Monday: [Day 6, Hex Nut](#)

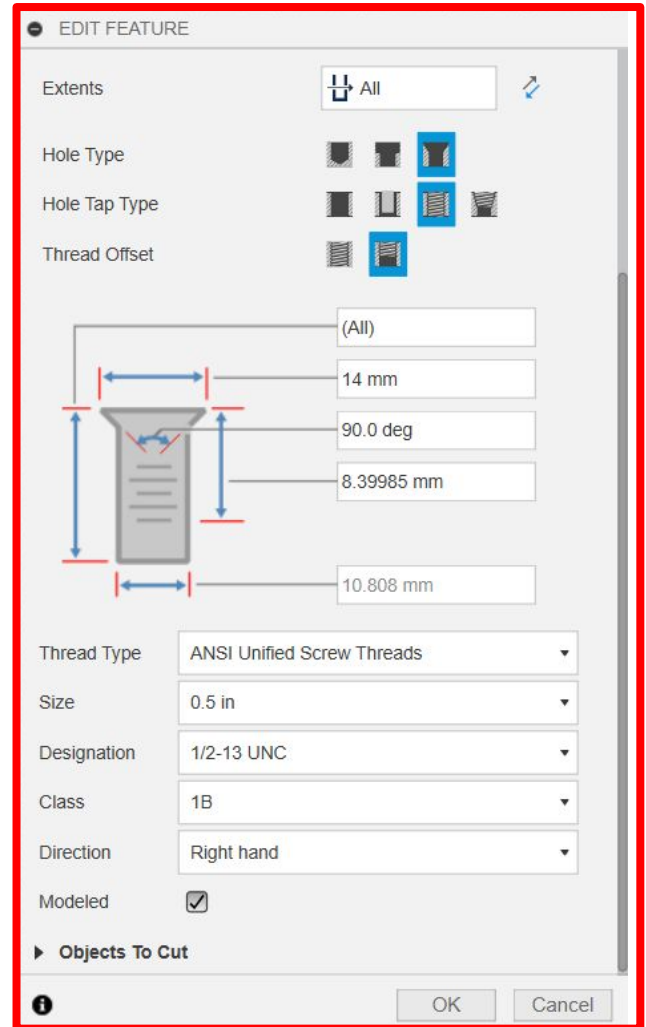
Tuesday: 3D Print 1/2" Hex Nut



Wednesday: [Day 7, Bicycle Grip](#)

Thursday: [Day 8, Door Stop](#), Learning Check

Friday: Shop/Classroom 5S



Tutorial Videos Week of 9-29-25

[Fusion 360 in 30 Days Video Tutorial Series](#)

Monday: [Day 9, Light Bulb](#)

Tuesday: [Day 10, Phone Case](#)

Wednesday: [Day 11, Dog Bowl](#)

Thursday: [Day 13, Components and Assembly Structure](#)

When the bell rings, be prepared to start class by:



1. Be seated

2. Cell phone in pouch

3. Backpack stowed

4. Have Fusion 360 on your computer and start relevant work